



Technical guidelines

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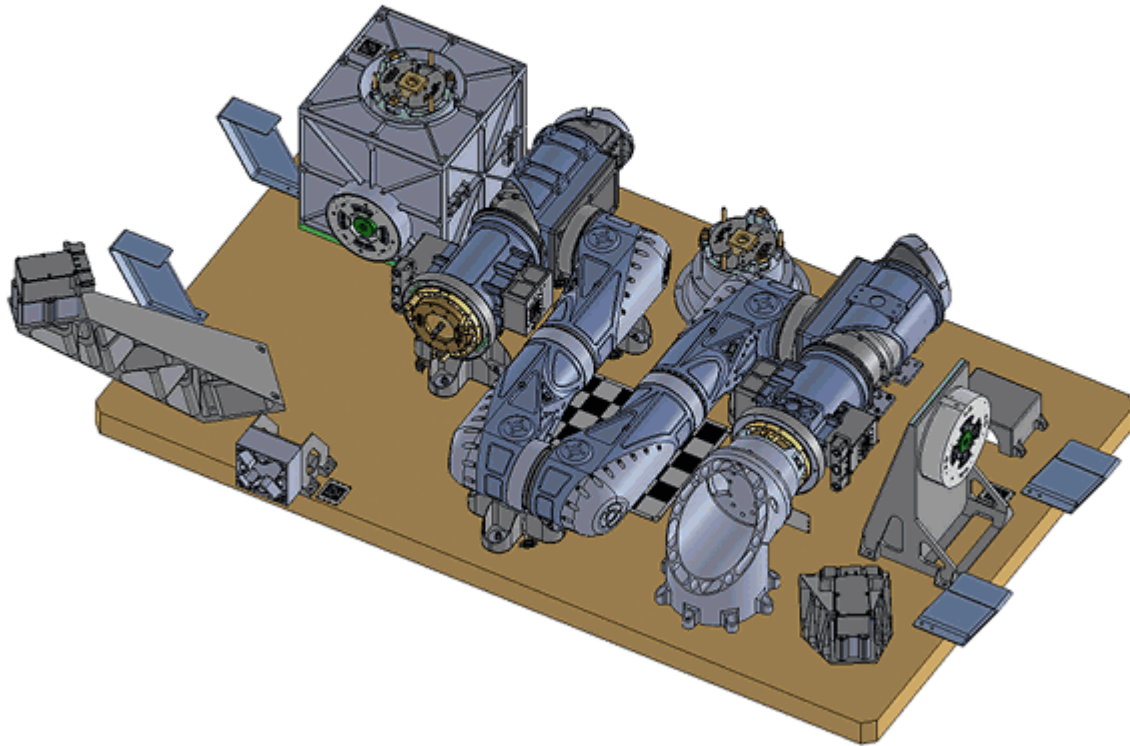
The FFR robotic arm

Payloads selected for this challenge will be manipulated by the robotic arm developed by Motiv Space Systems aboard the Fly Foundational Robots (FFR) spacecraft. The arm will be mounted to the FFR payload deck and capable of end-over-end walking and repositioning of Orbital Replacement Units (ORUs) in microgravity.

The arm's end effectors are active CrossLink cleats — one on each side — that can interface with passive CrossLink cleats mounted on the payload deck or on an ORU. During manipulation, the arm is capable of supplying power and data to a grappled ORU through its end effector.

The nominal concept of operations for a challenge payload is as follows: a delivery vehicle hosting the winning payloads from this challenge will rendezvous with the FFR spacecraft already in orbit; the robotic arm will transfer each payload from the delivery vehicle and install it onto the FFR platform via one of the available separable interfaces; the arm will then interact with, manipulate, or reconfigure the payload according to the demonstration plan developed by the team.

For more detailed diagrams and CAD models of the robotic arm and interfaces, please refer to the [technical overview document](#).



Examples of manipulation and potential demonstrations

The following examples are illustrative of the types of demonstrations that would be compatible with this challenge. This list is not exhaustive — applicants are encouraged to propose novel demonstrations that advance ISAM capabilities.

- **Robotic inspection and diagnostics:** a payload that, once installed on the FFR platform, deploys sensors or instruments that the arm can position, reorient, or activate to inspect spacecraft surfaces or components.
- **Modular payload swapping:** a payload designed to be detached from one interface, repositioned, and secured at another location on the deck, demonstrating on-orbit reconfiguration of modular systems.

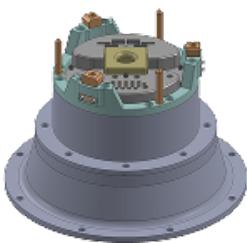
- **Structural assembly:** a payload that includes components the arm can manipulate and connect in orbit, demonstrating building-block approaches to in-space construction.
- **Sensor deployment:** a payload containing a folded or stowed instrument that the arm deploys, orients, or activates in orbit.
- **Material processing:** a payload that exposes materials or samples to the space environment and is repositioned or reconfigured by the arm at defined intervals.
- **Other novel demonstrations:** Applicants are encouraged to propose demonstrations not listed here, provided the concept advances ISAM capabilities and meets the technical requirements described in this document.

All demonstrations must be designed around the arm physically interacting with the payload — passively carrying an instrument from point A to point B does not constitute manipulation for the purposes of this challenge.

Payload form factor and requirements

1. Interface (mandatory)

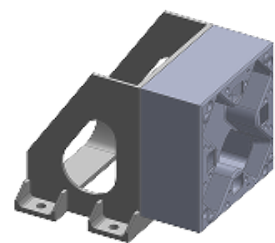
All payloads must incorporate a separable interface to enable mechanical docking to the FFR platform and manipulation by the robotic arm. Payloads must be designed to the mating interface specifications published in the [FFR Payload Separable Interfaces document](#). NASA will provide one of the interfaces listed below to each winner. Teams may propose a preferred interface in their application; however, final assignment of an interface will be made at the beginning of Phase 2 based on the team's preference combined with NASA's assessment of proposed payload design and flight demonstration plan.



CrossLink passive cleat



ISSI active cleat & bracket



Fuseblox passive

CrossLink Cleat (passive)

- Deck interface dimensions: 172mm × 111mm
- Mating interface: active CrossLink Cleat (robotic arm end effector)
- Power available: 28V at 3A (RA power); 24–32V at 3A (heater power); chassis ground
- Data available: 2x RS422 channels; ESTOP
- Two passive CrossLink Cleat positions available on the deck (one is reserved for the arm at all times)

iSSI — intelligent Space System Interface (active)

- Deck interface dimensions: 145mm × 49mm
- Mating interface: iSSI passive or active interface
- Power available: 24V at 3A; 24–32V at 3A (heater power)
- Data available: Gigabit Ethernet
- One position available on the deck

FuseBlox (passive)

- Deck interface dimensions: 100mm × 100mm × 40mm
- Mating interface: FuseBlox active interface
- Power available: 24–32V unregulated at 3A
- Data available: Gigabit Ethernet
- One position available on the deck

Note that passive interfaces (CrossLink Cleat and FuseBlox) do not initiate docking — docking is driven by the mating active interface. Applicants should account for this in their proposed flight demonstration plans.

2. Size, weight, and power (SWaP)

Payload dimensions must fall within the maximum cross-sections defined for the proposed interface, as illustrated in the [FFR Payload Separable Interfaces document](#). Allowable height depends on the placement of the ORU passive CrossLink Cleat and its relation to the arm workspace. (Note: winning teams will confirm their configuration with the FFR team prior to finalizing their design.)

Please note: Additional details regarding maximum mass per payload, power draw limits, and thermal dissipation constraints are forthcoming.

3. Additional requirements

- Payloads must be designed to survive launch loads and the LEO thermal and radiation environment for the duration of the mission.
- Payloads must not present a debris hazard. Any deployable elements must be tethered or otherwise retained.
- Payloads must comply with all range safety and launch vehicle requirements applicable to the delivery vehicle.
- Applicants are responsible for ensuring their payload designs do not create interference with FFR spacecraft systems or other payloads.

Hosted orbital flight parameters

NASA intends to provide the opportunity for winning payloads to fly aboard an orbital spacecraft that will rendezvous with the FFR spacecraft. The FFR mission is expected to launch in late 2027, and the TechLeap payloads are slated to launch in early 2028 on a spacecraft that will rendezvous with FFR.

Please note: Parameters that describe the expected flight environment are forthcoming.

Civil space shortfalls

Applicants are encouraged to consider NASA's 2026 Civil Space Shortfalls, which highlight technology areas requiring further development to meet future exploration, science, and other mission needs. Several of these shortfalls are related to ISAM and may be relevant to this challenge, including:

- SF16: Perform surface and planetary mobility and cargo operations
- SF27: Emplace and maintain large, stable space platforms and observatories
- SF28: Enable advanced robotics, automation, and autonomy for space exploration
- SF31: Provide space technology demonstration environments
- SF32: Develop an affordable and resilient supply chain for space exploration

Review the [full list of 2026 Civil Space Shortfalls](#).



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